

Design Guidance

- (f) Development is to consider:
 - (i) Physical characteristics of the site;
 - (ii) Site context, such as adjacent buildings or structures affecting the site, relationship of the site to the street, identification of key features such as views and orientation;
 - (iii) Overshadowing caused by existing buildings;
 - (iv) The orientation of true solar north, and a range of 30 degrees east and 20 degrees west of true north;
 - (v) Trees on, or affecting the site, identifying location, type, size and condition; and
 - (vi) Prevailing seasonal winds, sun and shade characteristics.
- (g) Development is to be orientated to achieve optimum solar access to thermal mass in winter, and shade thermal mass in summer. To achieve this:
 - (i) Shade north and west facing windows from direct summer sun with external horizontal shading devices such as awnings, upper floor balconies, eaves and overhangs; and
 - (ii) Utilise vertical shading devices such as vertical louvres or fins on east and west facing windows that consider the oblique angles of the sun.
 - (iii) The use of trees and shrubs as an additional method of shading a surface or window is encouraged.
- (h) Development must not unduly impact upon the ability of surrounding properties to achieve passive design strategies and solar access.
- (i) Insulation is to be used in external walls and roofs to reduce heat escaping from a building in winter and to maintain a lower internal temperature in summer. Position internal walls and partitions to allow for any prevailing passage of air through the building.
- (j) Development is to utilize operable natural ventilation to evacuate heat from roof or underfloor cavities in summer, and to retain warmth in winter. Design for cross - ventilation or stack-ventilation where possible to minimise the use of mechanical ventilation.
- (k) The use of green roofs or walls to reduce heat absorption and provide thermal mass to a development is strongly encouraged. Refer to *Part B3 Landscaping, Biodiversity and Vegetation Preservation* for additional information.
- (l) The use of trees and vegetation as an additional method of shading a roof, window or surface is strongly encouraged.